

Establishing Requirements

Assistive Devices for those on the Spectrum

For this week's assignment we will be working on establishing the various requirements that we will need to consider for our Assistive Device. We will establish the various pre-requisites that are required for our device and go through each of them. They are as follow:

1. **Functional Requirement:** The target audience for our device will be people who are on the spectrum (ADHD, Autism, Schizophrenia, etc.). The primary functional requirement for our device is that it must be able to locate a person who is in distress or lost in a public setting. Since, our target audience are more prone to losing themselves. We must be able to accurately pinpoint and locate them precisely and fast.
2. **Data Requirements:** Since our device utilizes concepts of Global Positioning Systems. We will require a large array of data such as Latitude, Longitude, Time, Accuracy, Persistence, etc. Furthermore, all of these data will need to be relayed at a Real-Time setup as they need to be constantly updated in-order to provide Real-Time updates. Also since we the proposed device is a wear-able device we need to ensure that the components that are used to obtain these data is of appropriate size since we need to maintain the overall aesthetic of the device while maintaining the applications and operations of the device
3. **Environmental Requirements:** There is much to be considered for this section so we will through them by individually
 - **Physical Environment:** The device will be primarily use in a Natural Setting which includes it being exposed to a moderate amount of lighting, noise and dust. Furthermore, is expected that

tracking the footsteps maybe erratic and not confined to a set path or design.

- **Social Environment:** For the most part our data may be required to be compatible to both synchronous as asynchronous. As there is a possibility that the data maybe need to be shared when locating the victim when he is in distress while at other times it may not be required.
- **Organizational Environment:** This section needs to be given at most importance since User support is one area that needs to be at the top at any given time. Furthermore, access to the user support must also be easy to use and handle, which may include a good user interface and so on.
- **Technical Environment:** Since our devices will be wearable it must be compatible with the popular systems such as IOs, Android, etc. In order to provide a seamless performance.